



LLM vs. 傳統 ML:

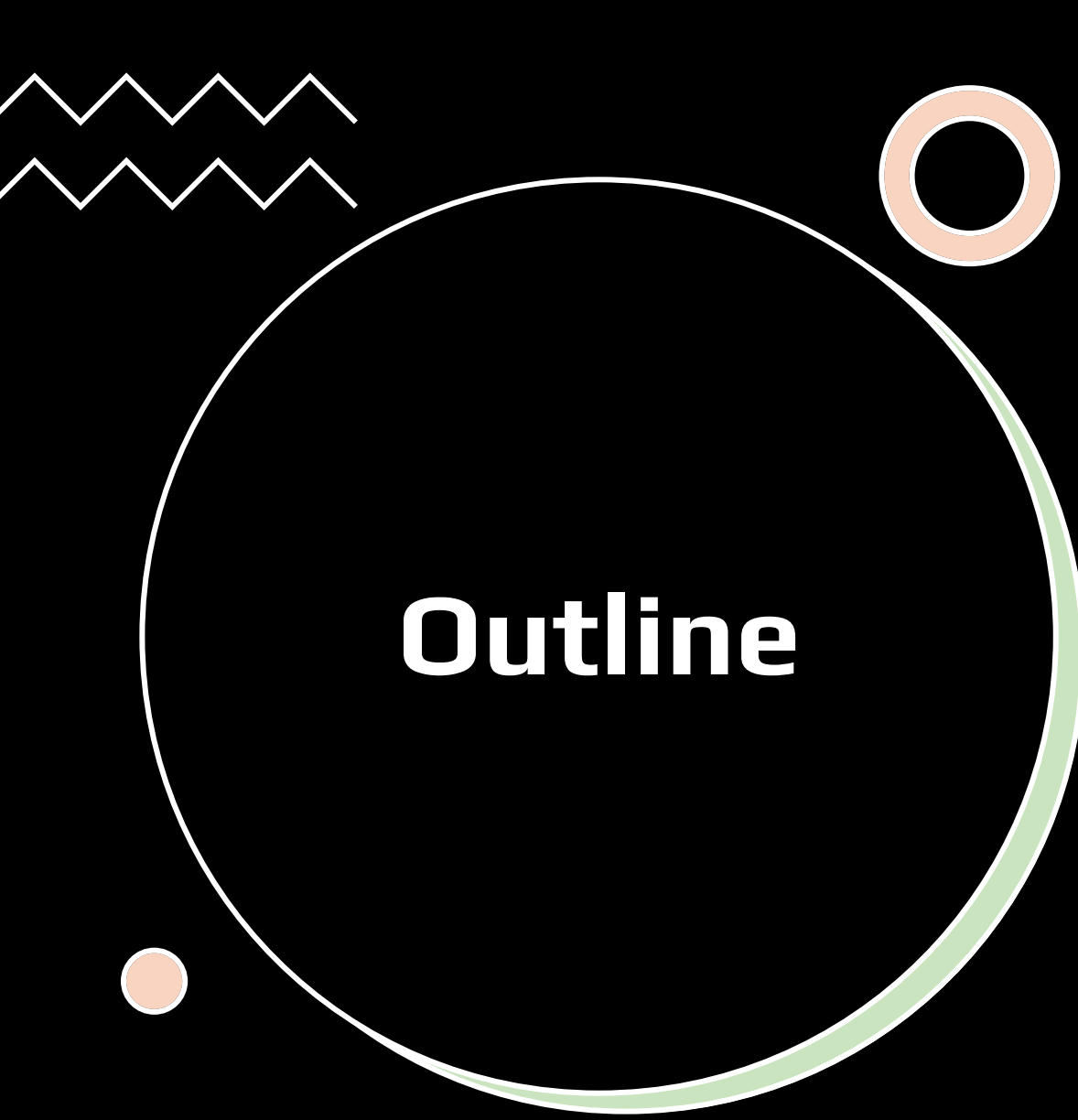
時序預測的跨域重編程

與效能權衡

參考論文: Time-LLM: Time Series Forecasting by Reprogramming
Large Language Models

論文網址: <https://arxiv.org/abs/2310.01728>

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Outline

- 核心痛點 (The Gap)
 - 解決方案 (The Solution)
 - 系統架構 (Architecture)
 - 實戰數據 (Performance)
 - 決策矩陣 (Trade-off)
 - 結論 (Conclusion)
- 

模型融合

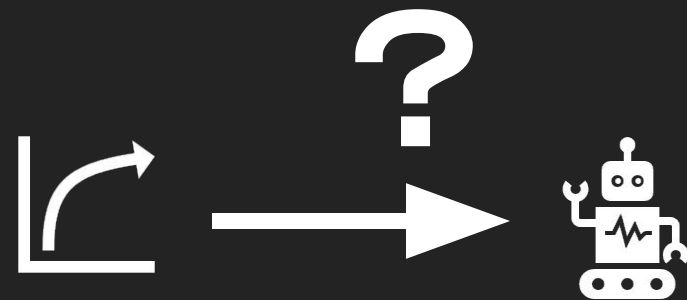
传统

informer, LSTM ; good for
number

LLM

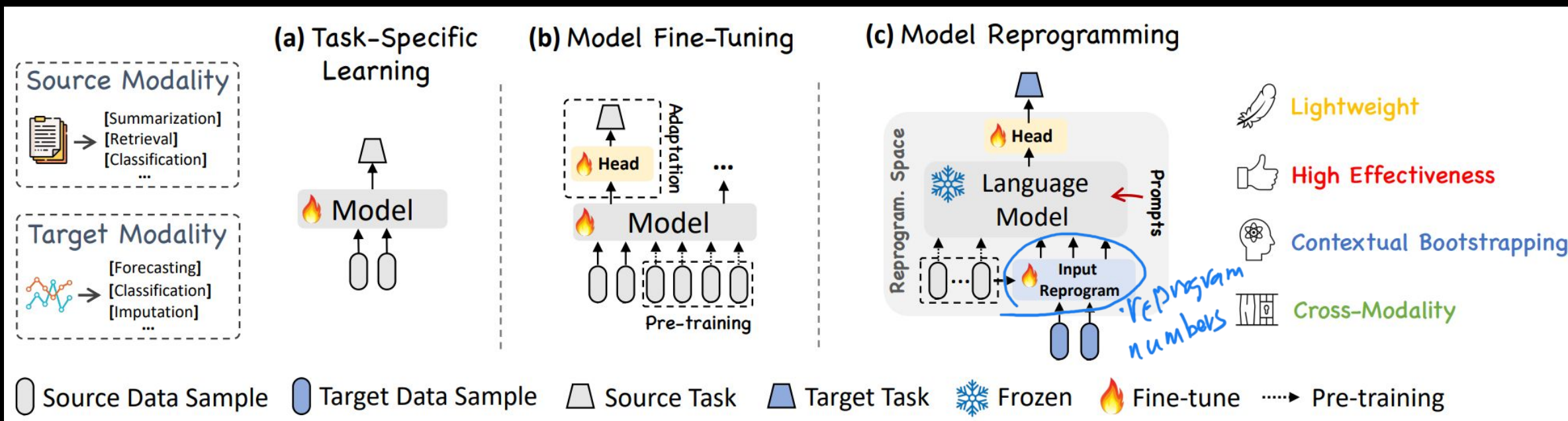
对数字不敏感
1024.56

integration



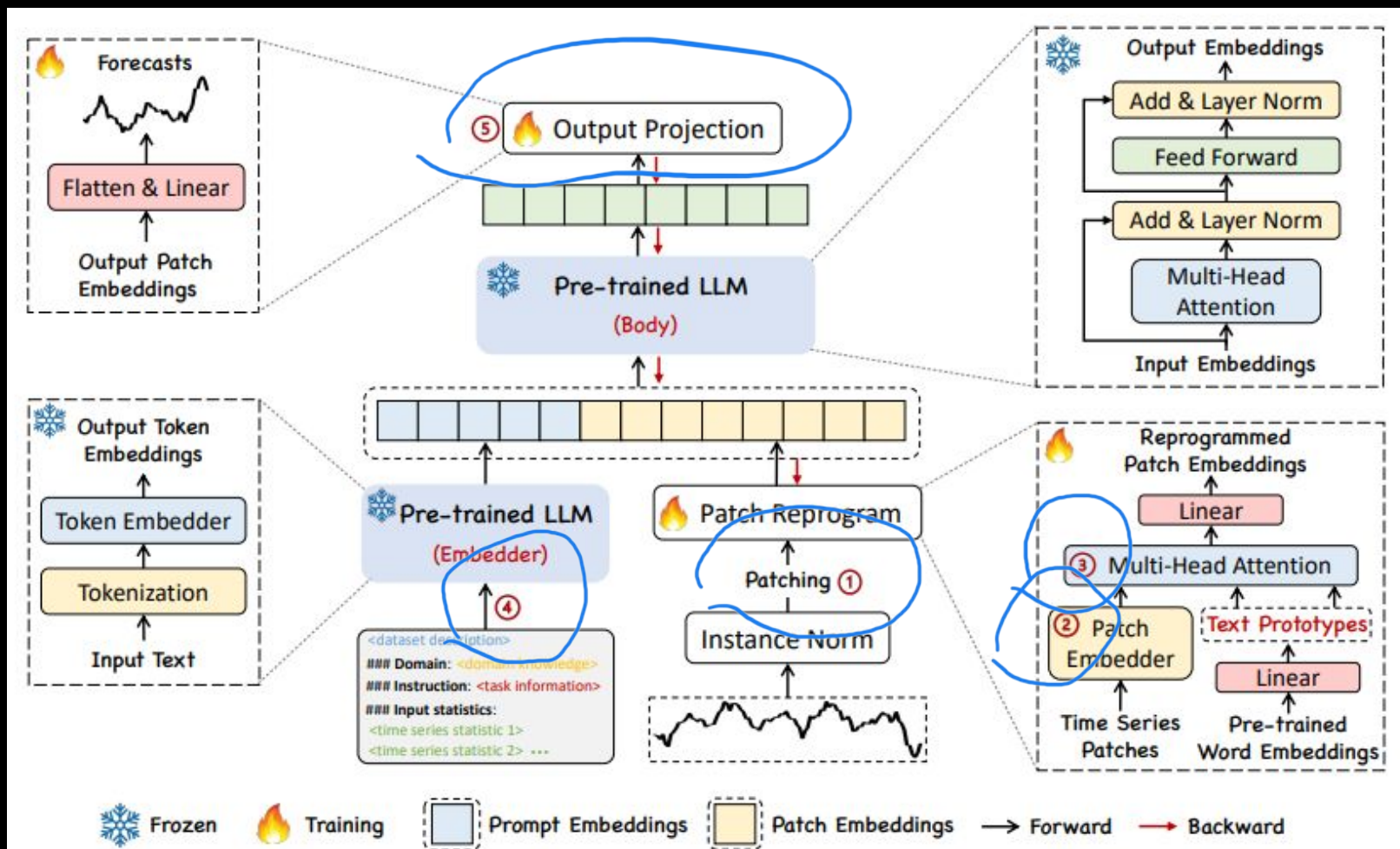


工程決策 - 微調還是重編程？





Time-LLM 系統架構





The Electricity Transformer Temperature (ETT) indicates the electric power long-term deployment. Each data point consists of the target oil temperature and 6 power load features ...
Below is the information about the input time series:

[BEGIN DATA]

[Domain]: We usually observe that electricity consumption peaks at noon, with a significant increase in transformer load

[Instruction]: Predict the next $\langle H \rangle$ steps given the previous $\langle T \rangle$ steps information attached

[Statistics]: The input has a minimum of $\langle \text{min_val} \rangle$, a maximum of $\langle \text{max_val} \rangle$, and a median of $\langle \text{median_val} \rangle$. The overall trend is $\langle \text{upward or downward} \rangle$. The top five lags are $\langle \text{lag_val} \rangle$.

[END DATA]

提示工程的創新

- Prompt-as-Prefix (PaP): 在輸入序列前加入自然語言的提示，增強推理能力。
- Prompt 三要素：
 - Dataset Context (資料集背景)
 - Task Instruction (任務指令)
 - Input Statistics (統計特徵)
- 優勢：將領域知識 (Domain Knowledge) 直接注入模型。



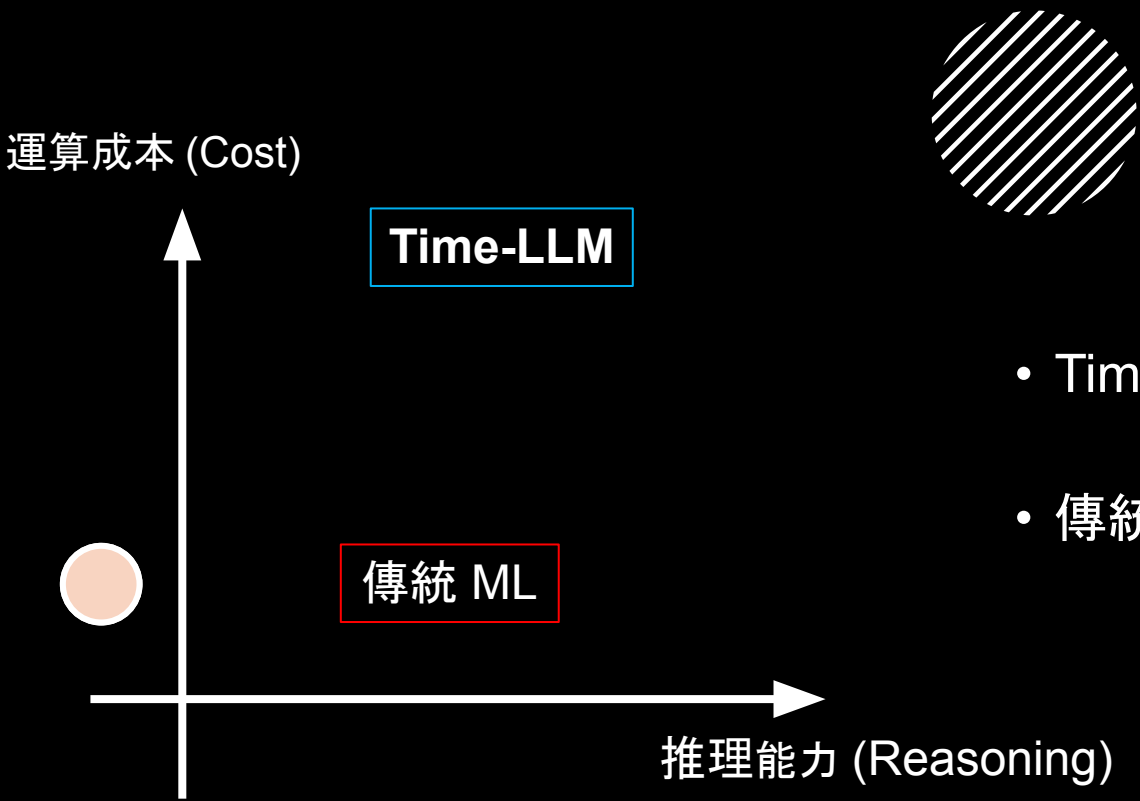
實戰數據 - 誰是預測王？

Methods	TIME-LLM (Ours)		GPT4TS (2023a)		DLinear (2023)		PatchTST (2023)		TimesNet (2023)		FEDformer (2022)		Autoformer (2021)		Stationary (2022)		ETSformer (2022)		LightTS (2022a)		Informer (2021)		Reformer (2020)	
Metric	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
ETTh1	0.556	0.522	<u>0.590</u>	<u>0.525</u>	0.691	0.600	0.633	0.542	0.869	0.628	0.639	0.561	0.702	0.596	0.915	0.639	1.180	0.834	1.375	0.877	1.199	0.809	1.249	0.833
ETTh2	0.370	0.394	<u>0.397</u>	<u>0.421</u>	0.605	0.538	0.415	0.431	0.479	0.465	0.466	0.475	0.488	0.499	0.462	0.455	0.894	0.713	2.655	1.160	3.872	1.513	3.485	1.486
ETTm1	0.404	0.427	0.464	0.441	<u>0.411</u>	<u>0.429</u>	0.501	0.466	0.677	0.537	0.722	0.605	0.802	0.628	0.797	0.578	0.980	0.714	0.971	0.705	1.192	0.821	1.426	0.856
ETTm2	0.277	0.323	<u>0.293</u>	<u>0.335</u>	0.316	0.368	0.296	0.343	0.320	0.353	0.463	0.488	1.342	0.930	0.332	0.366	0.447	0.487	0.987	0.756	3.370	1.440	3.978	1.587
Weather	0.234	0.273	<u>0.238</u>	<u>0.275</u>	0.241	0.283	0.242	0.279	0.279	0.301	0.284	0.324	0.300	0.342	0.318	0.323	0.318	0.360	0.289	0.322	0.597	0.495	0.546	0.469
ECL	0.175	<u>0.270</u>	<u>0.176</u>	0.269	0.180	0.280	0.180	0.273	0.323	0.392	0.346	0.427	0.431	0.478	0.444	0.480	0.660	0.617	0.441	0.489	1.195	0.891	0.965	0.768
Traffic	0.429	<u>0.306</u>	0.440	0.310	0.447	0.313	<u>0.430</u>	0.305	0.951	0.535	0.663	0.425	0.749	0.446	1.453	0.815	1.914	0.936	1.248	0.684	1.534	0.811	1.551	0.821
1 st Count	7		<u>1</u>		0		<u>1</u>		0		0		0		0		0		0		0		0	

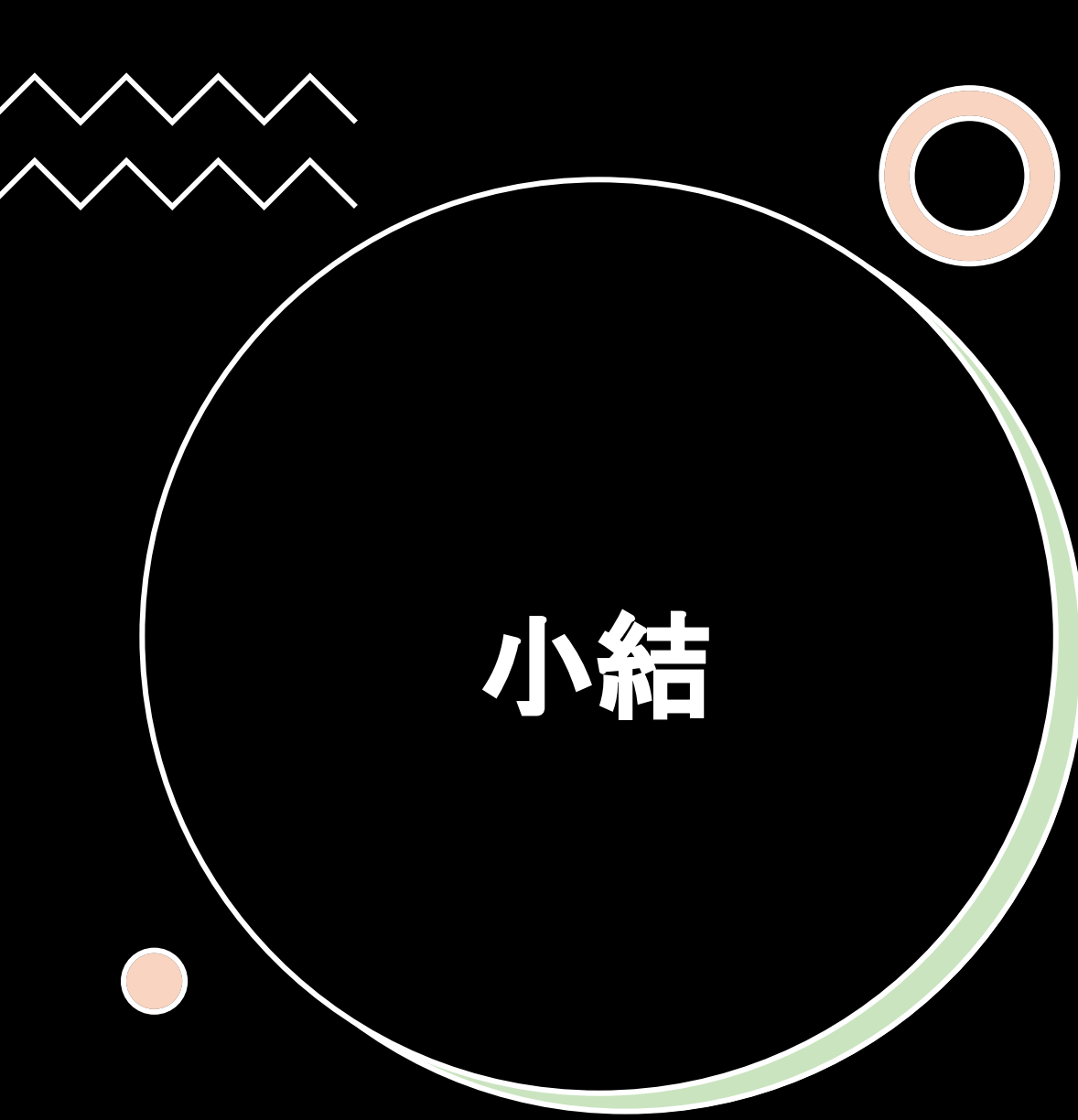




軟體工程師的決策矩陣



- Time-LLM (高成本/高推理): 適合宏觀分析、少樣本場景。
- 傳統 ML (低成本/低推理): 適合高頻交易、大數據量場景。



小結

1. 適配大於重練 (Adaptation > Retraining)

- 面對跨域難題, Time-LLM 證明了「重編程」比昂貴的「微調」更具工程效益。

2. 少樣本是殺手鐮 (Few-Shot is King)

- LLM 的最大價值不在於處理海量數據, 而在於對新興市場 (IPO) 或 冷門資產 的快速泛化能力。

3. 未來是混合架構 (The Future is Hybrid)

- 沒有萬能模型。最佳實務是讓 LLM 負責「語意推理」, 讓傳統模型負責「數值精算」。
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Thanks for your
attention